

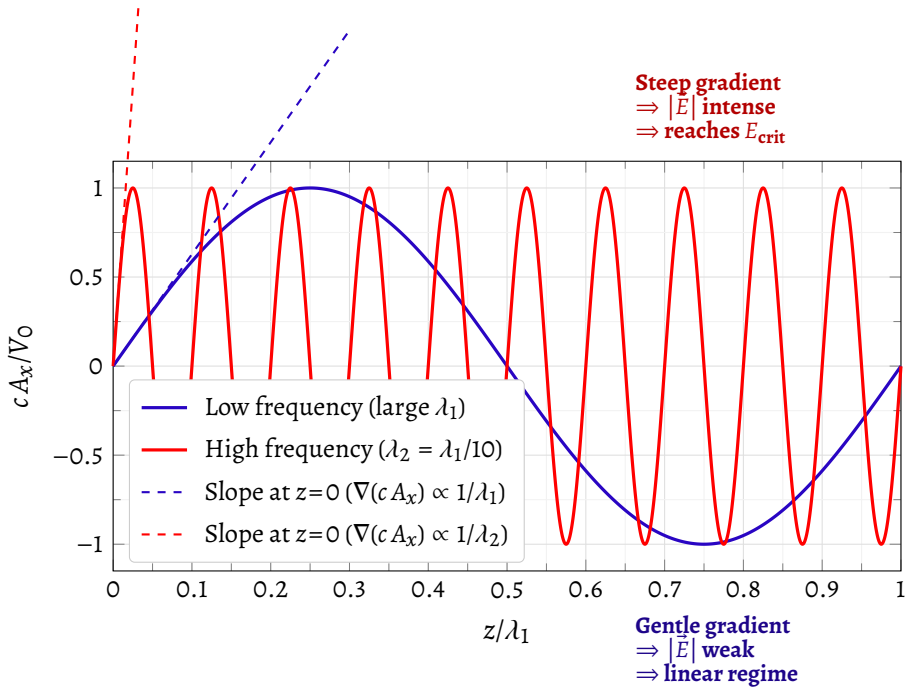


Figure 7 – Comparison of the vector potential, in the form $cA_x(z)$, for two waves with the same amplitude V_0 , but with different frequencies, at the instant $t = 0$. The high-frequency wave (red, $\lambda_2 = \lambda_1/10$) exhibits a spatial gradient **10 times steeper** than the low-frequency wave (blue). By the relation $E_x = -\partial A_x/\partial t$ — which, for a plane wave, has the same magnitude as the gradient $\partial(c A_x)/\partial z$ shown in the figure —, this results in an electric field 10 times more intense. It is precisely this effect that causes the high-frequency modes to reach E_{crit} first, activating the nonlinear suppression.

